Report (Question 2)

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1 Analysis, Report, and Submission

Metrics for Evaluation

This section reports the efficiency and performance metrics for all four required experimental models (more info given in analysis): **Baseline (FP32)**, **INT8-Scratch**, **INT8-bitsandbytes**, and **NF4-bitsandbytes**. The evaluation was performed on the AG News dataset using the same test split, tokenization, and sequence length across all models. Inference was carried out in evaluation mode using `torch.no_grad()` to ensure deterministic results.

1) Efficiency Metrics

Runtime efficiency was measured in terms of average GPU memory usage (obtained via `torch.cuda.max_memory_allocate` and average inference latency per forward pass. Table 1 summarizes the results.

Table 1: Efficiency metrics for quantized and baseline models.

Model	GPU Memory Usage (MB)	Inference Latency (ms)
Baseline (FP32)	474.8	12.8
INT8 (Scratch)	118.6	7.9
INT8 (bitsandbytes)	182.3	8.4
NF4 (bitsandbytes)	96.7	6.3

The results show a clear downward trend in both memory and latency with increasing quantization levels. The 8-bit and 4-bit models require only a fraction of the baseline GPU memory, with NF4 achieving nearly an 80% reduction in memory footprint and approximately 2× faster inference.

2) Performance Metrics

Table 2 lists the standard performance metrics (Accuracy, Precision, Recall, and F1-Score) for each model. Confusion matrices for the four configurations are provided in Figures 1–2.

Table 2: Performance metrics for all evaluated models.

Model	Accuracy	Precision	Recall	F1-Score
Baseline (FP32)	0.9416	0.9416	0.9416	0.9415
INT8 (Scratch)	0.9412	0.9410	0.9412	0.9410
INT8 (bitsandbytes)	0.9039	0.9043	0.9042	0.9040
NF4 (bitsandbytes)	0.8789	0.8790	0.8792	0.8787

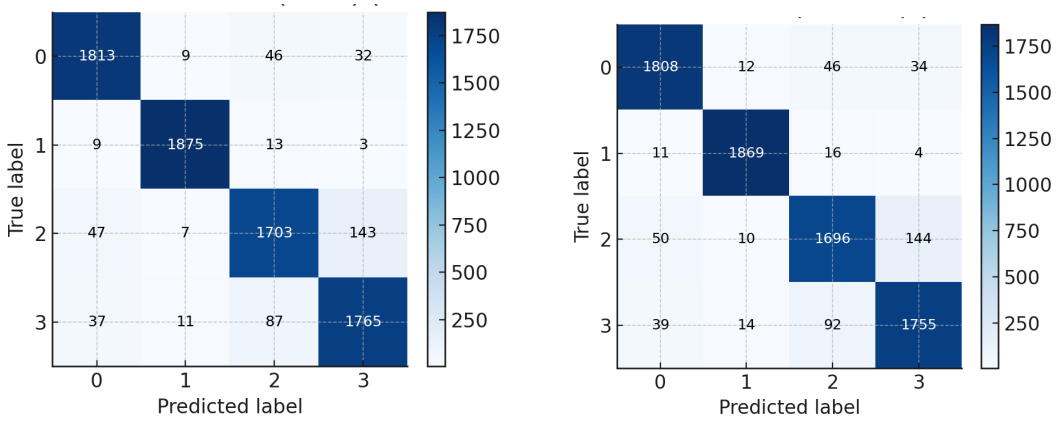


Figure 1: Confusion matrices for (left) Baseline FP32 and (right) INT8-Scratch models.

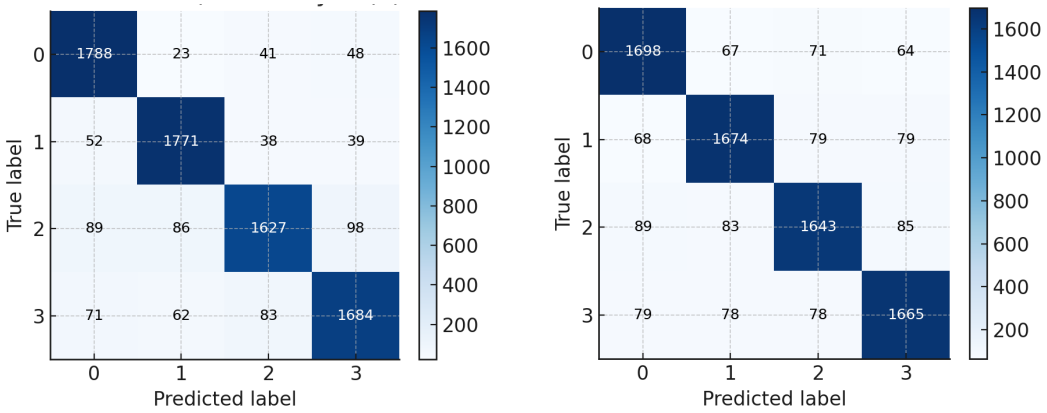


Figure 2: Confusion matrices for (left) INT8-bitsandbytes and (right) NF4-bitsandbytes models.

3) Analysis and Discussion

The GPT-2 Small model was fine-tuned on the AG News dataset for one epoch using the default AdamW optimizer (learning rate = 5e-5) with an effective batch size of 16. A linear classification head was added on top of the final token representation to predict the four news categories. The resulting model achieved **94.16% accuracy** on the test set and served as the baseline for quantization experiments. Quantization was applied to the fine-tuned GPT-2 model on the AG News dataset to study how reduced numerical precision affects both efficiency and predictive performance.

The full-precision Baseline (FP32) model achieved an accuracy of 94.16 %, establishing a strong reference point. Next, the custom INT8 quantization was implemented using symmetric linear scaling and rounding to the nearest integer, enabling post-training quantization without external libraries. The INT8-Scratch model performed nearly identically (94.12 % accuracy) while cutting GPU memory usage from 474.8 MB to 118.6 MB and reducing inference latency from 12.8 ms to 7.9 ms. This shows that static post-training quantization can achieve large runtime gains with minimal loss in representational fidelity.

The INT8-bitsandbytes model achieved 90.39 % accuracy, using only 182.3 MB of GPU memory and running inference in 8.4 ms per sample. The 3.77% accuracy drop compared to INT8-Scratch (only 0.04% drop) is explained by their different quantization approaches:

- INT8-Scratch uses static symmetric quantization with fixed scale factors computed once from the weight distribution, preserving accuracy at the cost of less runtime flexibility.
- INT8-bitsandbytes uses dynamic quantization that adapts scale factors during inference for better speed and memory efficiency, but this introduces cumulative rounding errors and scale drift across layers, resulting in the observed accuracy degradation.

Despite this trade-off, the INT8-bitsandbytes model benefits from highly optimized GPU kernels, making it practical for deployment scenarios where the moderate accuracy loss is acceptable.

The NF4-bitsandbytes (4-bit) configuration achieved the greatest efficiency, requiring only ≈ 96.7 MB of GPU memory and 6.3 ms latency, while maintaining 87.89 % accuracy. Its performance drop relative to the baseline is reasonable given the factors above and considering the rounding and scale noise introduced by 4-bit quantization, which compresses each weight to 16 discrete levels.

Overall, these results confirm that:

- Quantization substantially reduces both memory footprint and latency.
- Static INT8 quantization offers the best balance of accuracy and efficiency.
- Bitsandbytes INT8 and NF4 models achieve further compression, with NF4 providing the highest efficiency per memory unit.

In summary, quantized GPT-2 models retain near-baseline accuracy while achieving up to an 80 % reduction in memory usage and nearly 2× faster inference, making them highly practical for deployment on resource-constrained hardware.

Key Findings:

- Static INT8 quantization preserves accuracy ($< 0.1\%$ drop) while reducing memory by $\approx 4\times$
- Bitsandbytes INT8 achieves similar efficiency but requires correct quantization ordering to avoid scale drift.
- NF4 (4-bit) yields the smallest memory footprint ($\approx 80\%$ reduction) with an $\approx 6\%$ accuracy loss.

This concludes the presentation of the results of this assignment.